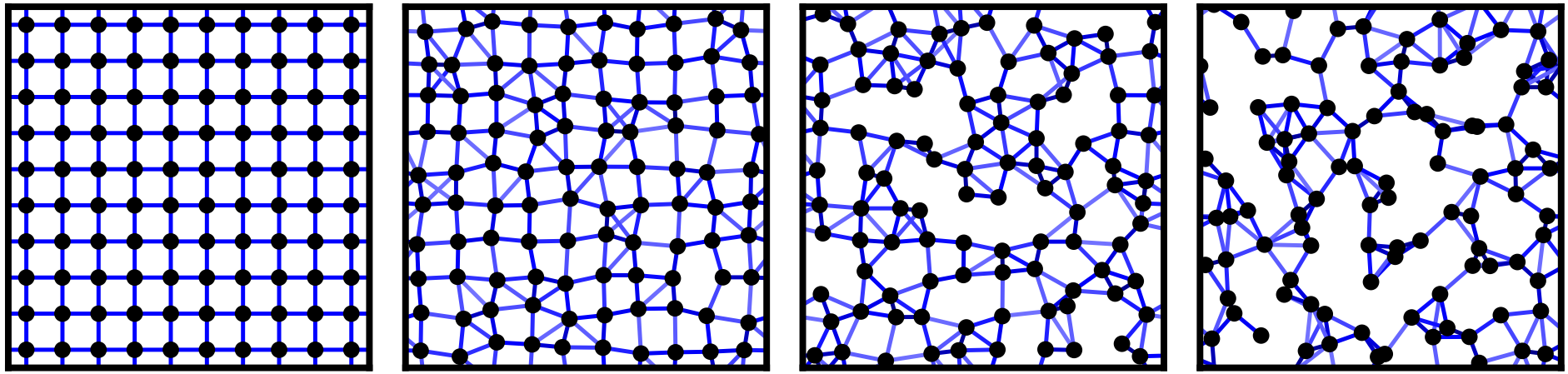
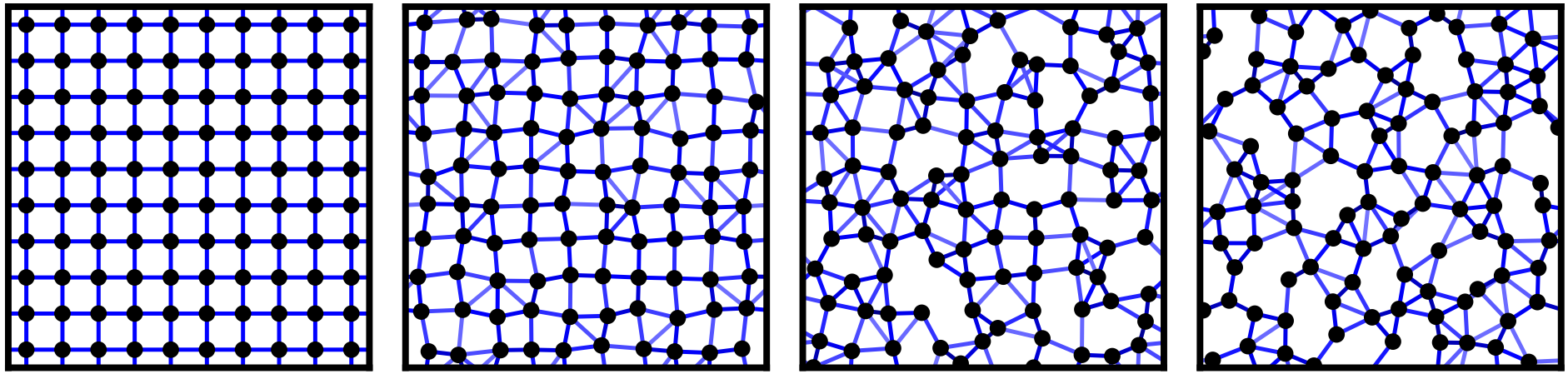
No repulsion, $v = 0$



Strong repulsion, $v = 0.3$



$\sigma = 0$ → $\sigma = 0.3\, r_0$